

# CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

Master of Technology (M.Tech.)

SEM - I University Examination, January - 2014

M.Tech. (CAD/CAM)

ME 703.01 Advanced Dynamics of Machines

Date: 04.01.2014, Saturday

Time: 10:00 a.m. to 01:00 p.m.

Maximum Marks: 70

## Instructions:

1. Figures to the right indicate *full* marks.
2. Make suitable assumptions and draw neat figures wherever if required.
3. *use separate answer book for each section.*

## SECTION- I

- Q1 (a) Synthesize and draw 4-bar mechanism to generate a function  $y = \log_{10}(x)$  in the range  $1 \leq x \leq 2$ , where  $\theta_1 = 60^\circ = \phi_1$  and  $\Delta\theta = 90^\circ, \Delta\phi = 60^\circ$ . Take three precision points and assume fixed link length as 40 mm 12

OR

- Q1 (a) Synthesis a slider crank mechanism for the following position of slider and crank. 12

$$S = 0.9215m, \theta = 30^\circ$$

$$\dot{S} = -2.23m/sec, \omega_2 = 20rad/sec$$

$$\ddot{S} = -83.84m/sec^2, \alpha_2 = 10rad/sec^2$$

Draw the mechanism by using Freudenstien's equation

- Q2 (a) For the static equilibrium of the quick-return mechanism shown in fig.1. Determine the input torque  $T_2$  to be applied on link AB for a force of 300 N on the slider D. The dimensions of the various links are OA = 400 mm, AB = 200 mm, OC = 800 mm, CD = 300 mm. Use Virtual work method. 06

- (b) Explain dynamical equivalent system and determine the correction couple to the system to reduce it to a dynamically equivalent system for the following: 06  
A connecting rod of an IC engine has a mass of 2kg and the distance between the centre of gudgeon pin and centre of crank pin is 250 mm. The C.G. falls at a point 100mm from the gudgeon pin along the line of centre. The radius of gyration about an axis through the C.G. perpendicular to the plane of rotation is 110mm. Find the equivalent dynamical system if only one of the masses is located at gudgeon pin. If the connecting rod is replaced by two masses, one at the gudgeon pin and the other at the crank pin and the angular acceleration of the rod is  $23000 rad/s^2$  clockwise.

OR

- Q2 (a) A four-link mechanism with the following dimensions is acted upon by a force  $80\angle 150^\circ N$  on link DC. AD=50 mm, AB=40 mm, BC=100 mm, DC=75 mm, DE=35mm. Determine the input torque T on the link AB for the static equilibrium of the mechanism for the given configuration use analytical method (fig.2). 08

- (b) Explain how we can consider static and turning friction in static force analysis. 04

P.T.O

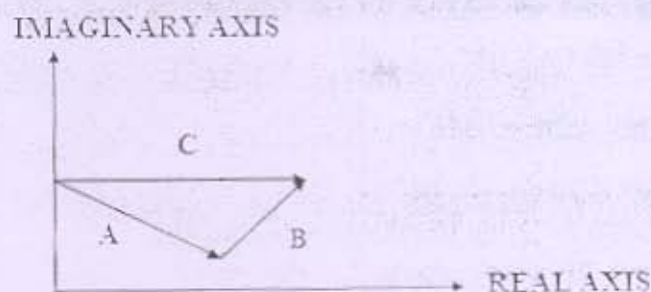
- Q3 (a) In 3-4-5 Polynomial cam follower rise 20 mm in  $120^\circ$  of cam rotation. The cam shaft rotates at 300 rpm. Find Displacement, Velocity, Acceleration and Jerk when shaft has turned through  $350^\circ$ . 05
- (b) Explain the jump Phenomenon. How will you determine jump speed for an eccentric cam operating a flat faced follower? 05

## SECTION - II

- Q4 ABCD is a four bar mechanism. Link length  $AB=50\text{mm}$ ,  $BC=90\text{mm}$ ,  $CD=110\text{mm}$ , Fixed link  $DA=125\text{mm}$  and  $\angle BAD = 50^\circ$ . If the link AB rotates at constant angular velocity of 12 rad/sec. Determine analytically, position and velocity of other links. 12

OR

- Q4 (a) Define following term (any two) 04
- 1) Partial degree of freedom
  - 2) Loop Closure equation
  - 3) Spatial Mechanism
- Q4 (b) Write the procedure to find i) Magnitude and direction of vector C ii) Magnitude of Vector B and vector C for the following figure. 08



- Q5 (a) Investigate the state of primary and secondary balancing of a four stroke cycle four cylinder engine with firing order I-II-III-IV. What will be the change in this state when the firing order is altered I-II-IV-III? 06
- (b) Investigate the state of unbalance of a two stroke radial engine with six cylinders by using Direct and Reverse Crank Method. 06
- OR
- Q5 (a) A three cylinder radial engine cylinders located  $120^\circ$  from neighboring cylinder. Reciprocating mass of each cylinder is 1.2 kg. Length of crank is 75 mm and each connecting rod is 250 mm long. Find out maximum primary and secondary unbalance forces if the engine runs at 2500 rpm. 06
- (b) Determine the primary and secondary forces for a V-engine if: 06
- i) Total angle of V is  $60^\circ$
  - ii) Total angle of V is  $90^\circ$
- Assume the crank position such that it bisects the line of action of the two positions.
- Q6 (a) What is Active balancing and passive balancing? Determine the favorable firing order for a Six- cylinder In-line engine. 04
- (b) Explain cam curvature and undercutting in roller followers 03
- (c) Derive a equation of Bloch's method of synthesis for 4-bar mechanism 04

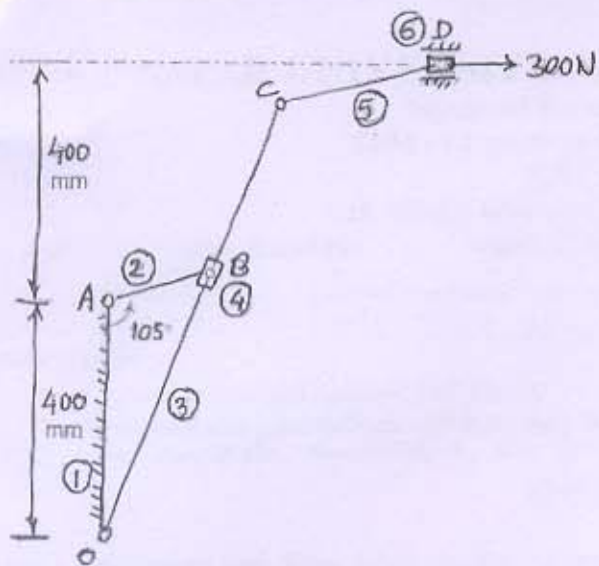


Fig.1

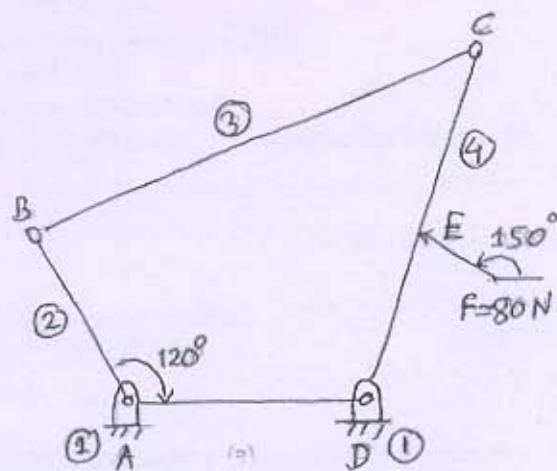


Fig.2